

Bellabeat Data Analytics Case Study Report

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Report: Bellabeat Fitness Tracker Analysis Case Study

1. Introduction

In this case study, I perform the real-world tasks of a junior data analyst working for Bellabeat, a high-tech manufacturer of health-focused smart products for women. To unlock new growth opportunities and answer the core business questions posed by executive leadership, I follow the structured phases of the data analysis process: **Ask, Prepare, Process, Analyze, Share, and Act**. This report serves as a comprehensive look at my data cleaning pipeline, key analytical findings, and data-driven marketing recommendations.

2. Background

Project Introduction

As a Junior Data Analyst on the Bellabeat marketing analytics team, I am conducting this case study to help guide the company's future global marketing strategy. By analyzing non-Bellabeat smart device fitness data, I aim to uncover deep consumer behavior trends and apply these insights to help Bellabeat scale up its market presence.

Business Scenario

The cofounder and Chief Creative Officer of Bellabeat, Urška Sršen, believes that analyzing consumer smart device data can unlock massive growth opportunities. My primary objective is to analyze how consumers use their smart devices to understand their daily activity, sleep, and energy dynamics. These insights will guide the marketing team in refining product features and crafting targeted digital campaigns.

Organizational Context

Founded in 2013, Bellabeat has rapidly grown into a tech-driven wellness company empowering women with knowledge about their own health and habits. Bellabeat distinguishes itself through beautifully designed, female-focused technology. The ecosystem includes:

- **Bellabeat App:** Provides users with health data related to activity, sleep, stress, and mindfulness.

- **Leaf & Time:** Classic wellness trackers and timepieces monitoring user activity and sleep.
- **Spring:** A smart water bottle tracking daily hydration levels.
- **Bellabeat Membership:** A subscription program offering personalized guidance on nutrition, activity, and sleep.

3. Data Analysis Processes

Phase 1: Ask

Business Task

Analyze public non-Bellabeat smart device usage habits to uncover behavioral trends and provide high-level, data-backed marketing recommendations for the Bellabeat product line.

Analysis Questions

1. What are some trends in smart device usage?
2. How could these trends apply to Bellabeat customers?
3. How could these trends help influence Bellabeat marketing strategy?

Phase 2: Prepare

Data Source

The primary data source utilized is the **FitBit Fitness Tracker Data** (CC0: Public Domain, made available by Mobius on Kaggle). This dataset contains personal fitness tracker outputs from eligible Fitbit users who consented to submit minute-level data, including physical activity, sleep monitoring, and METs. The collection period covers a 31-day window, capturing localized daily habits.

Data Limitations

A major limitation noted is the small sample size (33 unique user IDs represented across main tables) and the lack of demographic or gender-specific information. To offset these tracking scope limitations, the data was thoroughly aggregated to focus heavily on broad behavioral trends across days of the week.

Data Organization

The data was collected across multiple tables. Key variables and structural definitions map out as follows:

No.	Variable	Description
1	Id	Unique anonymized user identification key
2	Day Of Week	Extracted text indicator for the specific day of the week
3	Total Steps	Total step count accumulated by the user
4	Total Distance	Total distance covered by the user
5	Total Calories	Total energy burned in calories
6	Total Mets Sum	Accumulated Metabolic Equivalent of Task values tracking physical effort
7	Avg Sleep Time Hour	Actual duration spent dead asleep in hours
8	Avg Time Bed Hour	Total hours spent in bed (sleeping + awake time)
9	Avg Wasted Bed Time Min	Calculated minutes spent awake/restless in bed
10	User Type	Categorical classification based on user step volume

Phase 3: Process (Data Cleaning & Transformation)

To ensure high data integrity, strict analytical consistency, and dashboard readiness, all datasets were processed using SQL Server Management Studio (SSMS).

Key Processing Steps:

1. **Handling Incompleteness:** Removed rows with Null or structural zero values across key activity metrics to prevent artificial skewing.
2. **Date and Aggregation Alignment:** Extracted weekday classifications (Day Of Week) to map group behavioral patterns systematically across the 7-day cycle.
3. **Calculated Field Engineering:** Formulated the calculated field Avg Wasted Bed Time Min by isolating the exact duration difference: $(\text{Avg Time Bed Hour} - \text{Avg Sleep Time Hour}) * 60$.
4. **Data De-duplication:** Filtered out duplicate entry records across daily sleep logs to establish clean, singular daily averages per user ID.
5. **User Segmentation:** Categorized users into distinct behavioral profiles (Sedentary, Lightly Active, Fairly Active, Very Active) based on baseline daily step thresholds.

Phase 4: Analyze

The processed, clean tables were imported into Tableau to construct interactive, story-driven visualizations (figures) and dashboards. The findings are organized below.

Dashboard 1 Visualizations:

Figure 1: Distribution of Total Study Participants across Active Lifestyle Tiers

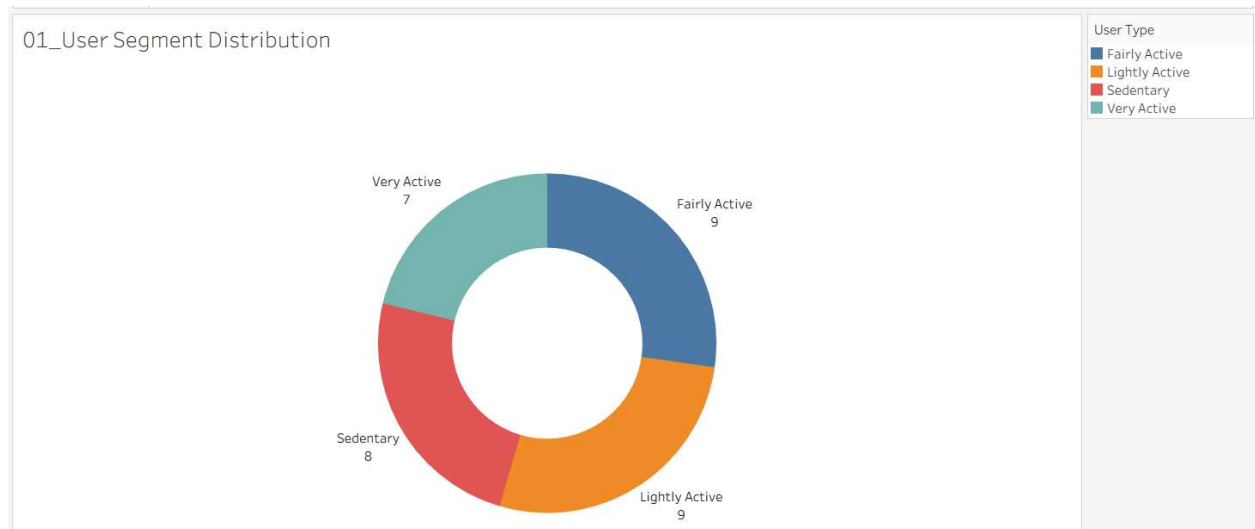


Figure 1 is a clean donut chart that categorizes the high-level user demographic of the study sample based on their recorded activity levels. The total participant group (n = 33) is split into four distinct, color-coded behavioral segments derived from their daily movement volume: **Fairly Active** (9 users), **Lightly Active** (9 users), **Sedentary** (8 users), and **Very Active** (7 users).

Key Analytical Insights:

- **Dominance of Lower-Activity Tiers:** Over half of the tracked user base (nearly **51.5%**) falls into the *Sedentary* or *Lightly Active* categories. This establishes that the majority of the target market struggles to maintain robust baseline movement throughout the day.
- **Product Strategy Implication:** Since the largest clusters are concentrated in the lower-activity zones, Bellabeat does not need to market heavily to ultra-marathoners or elite fitness enthusiasts. Instead, the primary customer acquisition and retention opportunity lies in providing accessible scaffolding, features, and guidance to help the sedentary and lightly active majorities gradually build healthier daily habits.

Figure 2: Weekday Volume Trends in Daily Step Counts Across the Study Group

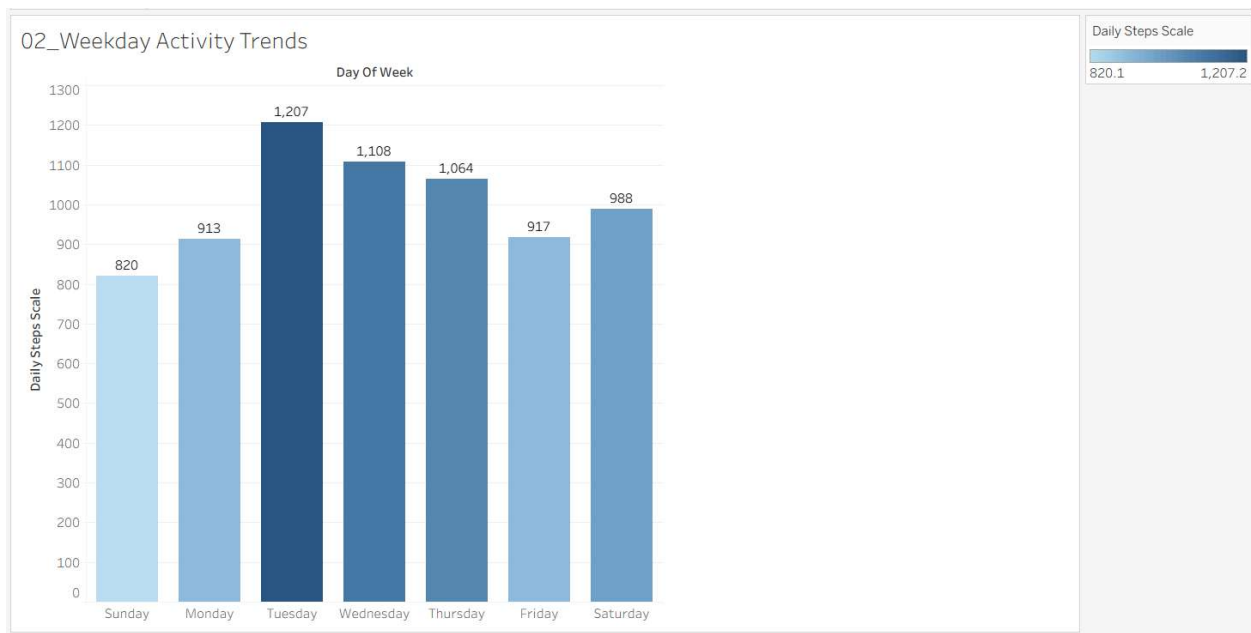


Figure 2 is a sorted column chart illustrating the fluctuating patterns of physical activity—measured in average daily steps—across different days of the week for the tracked consumer base. The bars are dynamically colored based on volume intensity (utilizing the Daily Steps Scale gradient) and are clearly labeled with their respective

integer values: **Tuesday** (1,207), **Wednesday** (1,108), **Thursday** (1,064), **Saturday** (988), **Friday** (917), **Monday** (913), and **Sunday** (820).

Key Analytical Insights:

- **The Mid-Week Peak:** Physical activity aggressively concentrates in the middle of the work week, hitting its absolute maximum on **Tuesday** (1,207 steps). This indicates that users maintain a more disciplined, active routine during their structured weekly commitments.
- **The Weekend Slump:** Contrary to the common assumption that people exercise more during their free time, activity trends downward into the weekend, bottoming out severely on **Sunday** (820 steps).
- **Product Strategy Implication:** This clear behavioral pattern exposes a major engagement vulnerability. Bellabeat can leverage these timing dynamics by designing localized smart notifications. Pushing tailored motivational cues or weekend walking challenges through the app on Friday afternoons can directly counter this recurring weekend drop-off, helping users smooth out their consistency gap.

Figure 3: Combined Analysis of Cumulative Group Distance Covered and Caloric Burn by Weekday

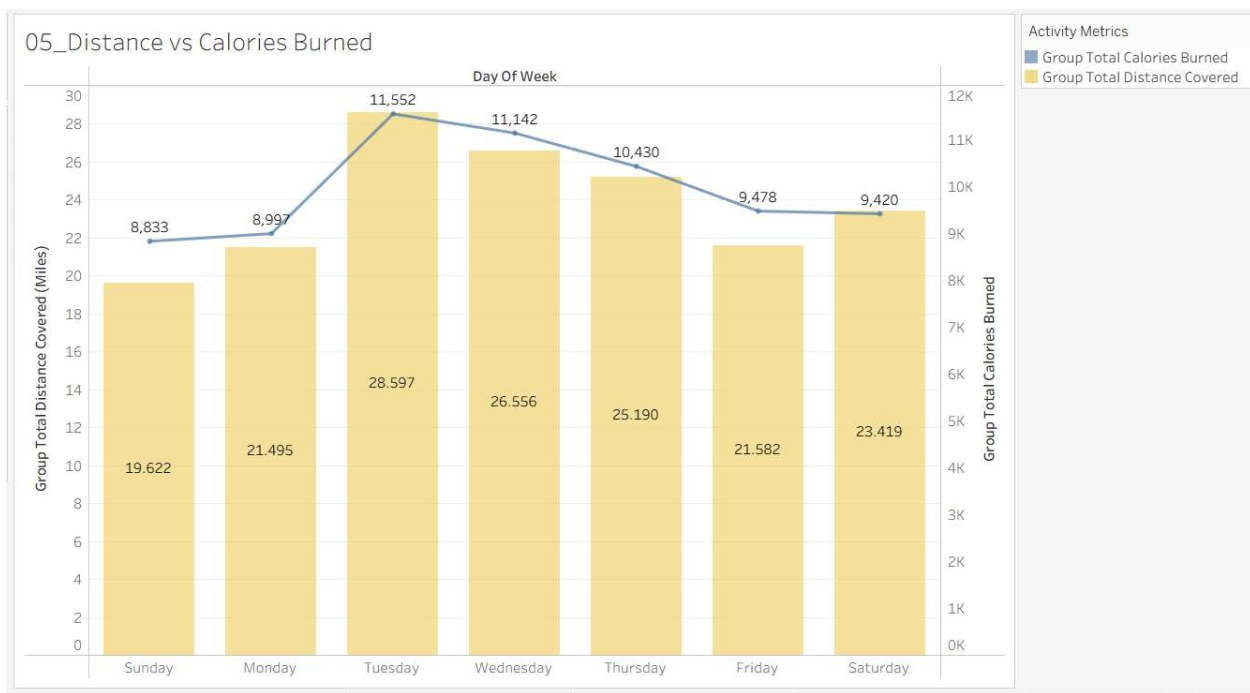


Figure 3 is a dual-axis combination chart that was constructed by aggregating the daily volume metrics of the study group across the tracking period to evaluate the direct

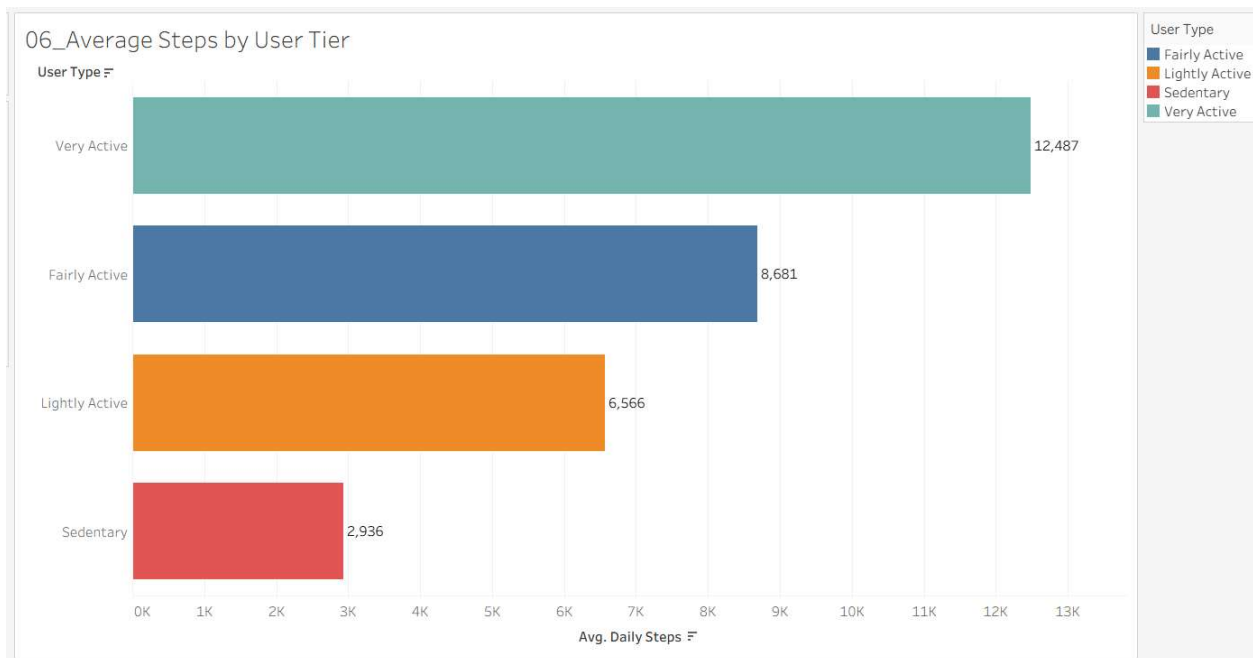
relationship between movement and energy expenditure. The horizontal axis displays the chronological flow of the week from Sunday to Saturday.

The vertical axis on the left tracks the cumulative distance covered by the study group in **Miles**, while the secondary vertical axis on the right measures the cumulative group **Calories Burned**. The visualization uses a line graph overlaying structured bar marks to assess how group movement volume correlates with total metabolic energy expenditure across the 31-day study window.

Key Analytical Insights:

- **Flawless Volumetric Covariance:** The visualization displays a direct, parallel covariance between physical displacement metrics and energy output. When the group's collective distance peaks on **Tuesday at 28.59 cumulative miles**, the total group energy expenditure hits its corresponding maximum at **11,552 total calories burned**.
- **The Friday Calorie Divergence:** A noticeable variance occurs on **Friday**. While the group's collective distance covered drops significantly to **21.58 miles** (closely matching Sunday's low tracking volume), the caloric expenditure remains resiliently elevated at **9,478 total calories**. This structural divergence indicates that on Fridays, participants engage in stationary, high-exertion physical activities (such as resistance weight training, circuit training, or stationary cycling) that burn substantial calories without accumulating forward travel distance. As Friday evenings mark the beginning of the weekend, it is assumed that more people have the time for stationary activities carried out in places such as gyms or training centers. Similar trends continue into Saturday and Sunday as these 2 days are also a part of the weekend. **Sunday** records the **least distance** covered at **19.62 miles** and also the **least number of calories burned** at **8,833**.
- **Product Strategy Implication:** This insight proves that raw mileage or step counts do not tell the full story of consumer health. Bellabeat should center its marketing messaging on the comprehensive tracking capabilities of its ecosystem. By showcasing how the *Bellabeat App* syncs with hardware devices to accurately credit women for high-intensity calorie burns during non-distance activities (such as yoga, pilates, or strength training), the company can position its products as true holistic wellness companions rather than basic pedometers.

Figure 4: Comparative Breakdown of Average Daily Steps by Active Lifestyle Tiers



This sorted horizontal bar chart contrasts the stark performance gaps between the four user lifestyle segments established in the participant demographic distribution. The vertical axis organizes the consumer segments cleanly from highest to lowest volume output, while the horizontal axis measures the baseline **Average Daily Steps**. Each bar is clearly labeled on its inside right margin with its precise calculated average value: **Very Active** (12,487 steps), **Fairly Active** (8,681 steps), **Lightly Active** (6,566 steps), and **Sedentary** (2,936 steps).

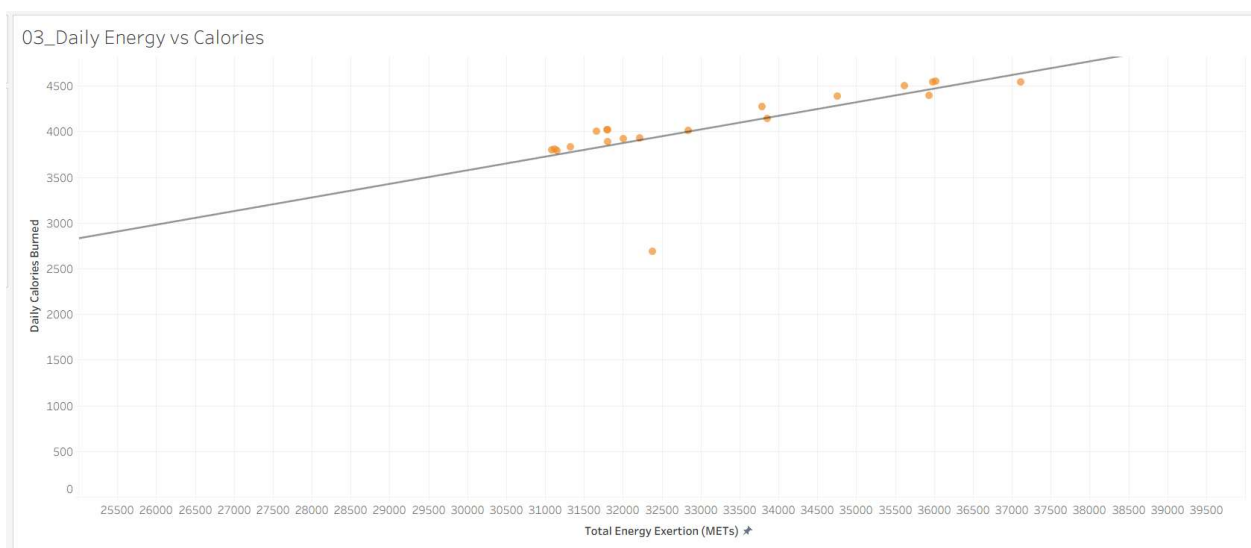
Key Analytical Insights:

- **The Performance Chasm:** The visualization exposes an aggressive **325% performance drop** between the *Very Active* tier and the *Sedentary* tier. Sedentary users are failing to meet minimum standard clinical health recommendations for daily movement (10,000 steps), averaging under 3,000 steps per day.
- **The High-Potential Target:** The *Lightly Active* (6,566 steps) and *Fairly Active* (8,681 steps) segments represent a massive "bridge" audience. These users are already wearing their smart devices and exhibiting consistent baseline movement habits, meaning they do not require a massive lifestyle overhaul—only minor behavioral adjustments—to cross the threshold into optimal daily activity.
- **Product Strategy Implication:** This metric highlights that a "one-size-fits-all" fitness goal is intimidating and ineffective for a diverse consumer base. Bellabeat

should use these insights to market personalized, tiered milestone achievements within its app interface. Rather than pushing a sedentary user to immediately aim for 10,000 steps, the app should reward incremental progression (for example: leveling up from the Sedentary baseline to a Lightly Active 5,000 step milestone), gamifying the consumer journey to build long-term user retention and product loyalty.

Dashboard 2 Visualizations:

Figure 5: Physiological Relationship Between Total Energy Exertion (METs) and Daily Calories Burned



This scatter plot maps the mathematical relationship between physical effort and metabolic output across the study group. The horizontal axis measures **Total Energy Exertion (METs)**, custom-scaled from 25,000 to 40,000 to eliminate empty chart space and focus entirely on the active data cluster. The vertical axis tracks **Daily Calories Burned**, ranging from 0 to 4,500. A solid, upward-sloping linear regression trend line is overlaid across the individual user data points to model the predictability of the correlation.

Key Analytical Insights:

- **Unmistakable Positive Correlation:** The visualization demonstrates an undeniable, direct positive correlation between systemic physical exertion and caloric expenditure. As a user's accumulated MET score increases along the horizontal axis, their daily calorie burn climbs reliably in tandem.
- **The Variance of True Exertion:** Zooming the axis into the 25,000 to 40,000 MET range exposes the true variance in user behavior. It highlights that the most

active individuals are consistently burning between 3,500 and 4,000 calories per day, driven entirely by sustained physical effort rather than baseline resting metabolism alone. It also clearly flags a single low-performing outlier who, despite accumulating over 32,000 METs, experienced a significantly lower caloric burn of roughly 2,600 calories, indicating potential individual baseline variations.

- **Product Strategy Implication:** This chart provides a powerful talking point for Bellabeat's marketing and software engineering teams. It mathematically proves that the tracker's internal proprietary algorithms are accurately anchoring calorie metrics to verified physiological exertion (METs). Bellabeat should feature this predictive accuracy when marketing its premium *Membership* platform. By showcasing how the app translates hard hardware tracker data into clear, scientifically sound metabolic insights, Bellabeat can build trust with women who want an honest, highly accurate view of their daily energy expenditure and workout efficiency.

Figure 6: Comparative Analysis of Total Time Spent in Bed vs. Actual Hours Slept



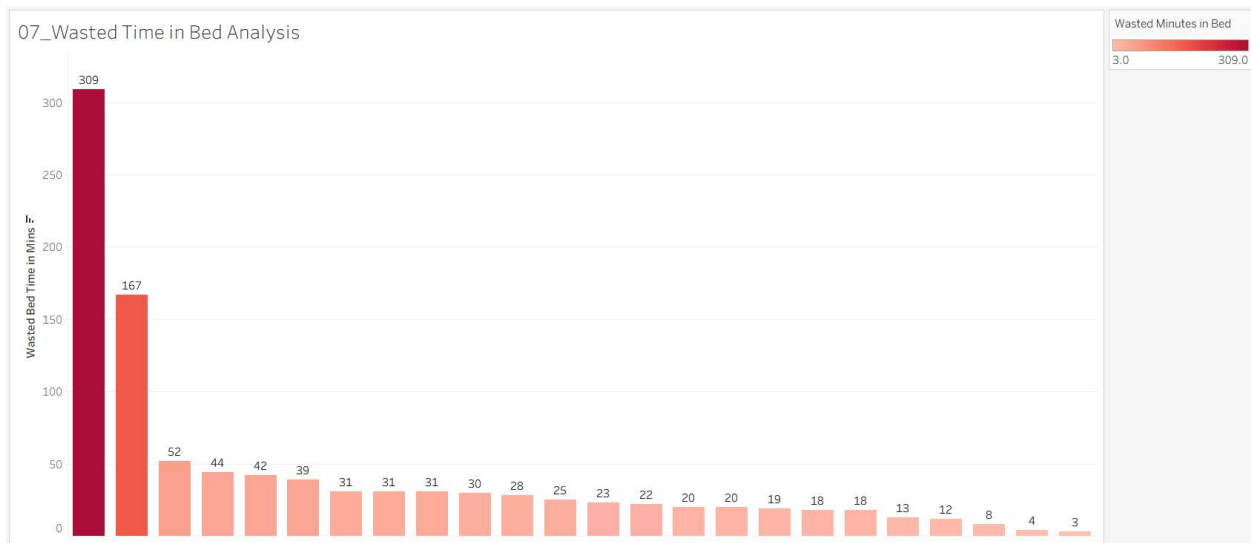
This nested thermometer-style bar chart illustrates the relationship between behavioral sleep intent and actual physiological rest for individual participants across the study group. The vertical axis measures duration in hours (scaled from 0 hours to 16 hours), while the horizontal axis displays unique tracking records across the month. Here, I have taken into account the average time spent in bed (Hours) along with the average estimate of actual hours slept. For each tracking record, a narrow blue bar (**Actual Hours Slept**) is nested directly inside a wider orange bar (**Time Spent in Bed**), creating an immediate visual indicator of individual sleep efficiency.

Note: Data labels were intentionally excluded from this visualization to maintain a clean, high-density layout and prevent text overlapping. Instead of tracking exact numerical values for every record, the visualization relies on the **relative height difference** between the nested bars to tell the story. Whenever an orange bar towers significantly over its nested blue bar, it visually quantifies an instance of poor sleep efficiency.

Key Analytical Insights:

- **The Persistent Structural Gap:** The chart exposes a widespread, systemic gap across nearly all tracking records. While users consistently dedicate an optimal 7 to 9 hours of time to being in bed, their actual hours of sleep frequently drop below 6 hours, and in severe instances, plummet to under 4 hours.
- **The Boundless Sleep Anomaly:** A prominent behavioral outlier is captured on the far left of the chart. This individual spent an extraordinary **16 total hours in bed**, yet only achieved nearly **11 hours of actual sleep**. This proves that simply extending time spent in bed does not automatically result in proportional, restorative rest.
- **Product Strategy Implication:** This structural insight provides a clear roadmap for Bellabeat's marketing and software engineering teams. The data proves that customers do not necessarily need to be told to *go to bed earlier*; rather, they need active support *optimizing the time they already spend in bed*. Bellabeat should use this finding to promote in-app sleep hygiene features, such as automated bedroom wind-down reminders, guided breathing exercises, or smart alarms that monitor sleep cycles. Bellabeat should position its ecosystem as an active tool for sleep optimization rather than a passive data logger.

Figure 7: Pareto Distribution of Average Wasted Time in Bed Across Individual Users



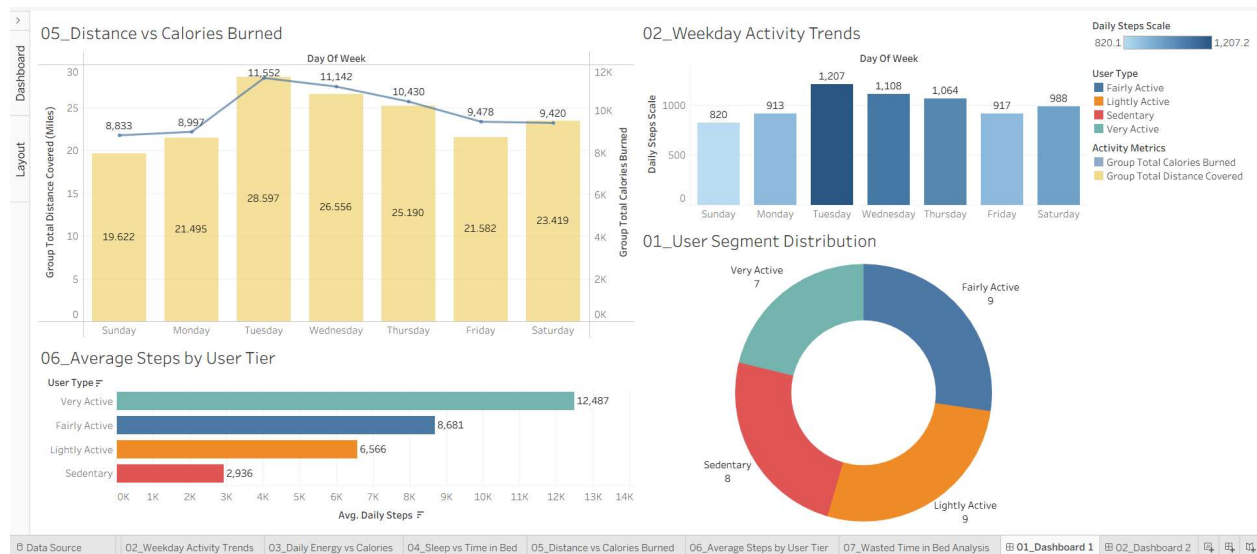
This ranked column chart isolates and quantifies sleep inefficiency by tracking the average daily minutes users spend awake, restless, or unable to fall asleep while lying in bed. The vertical axis measures **Avg Wasted Bed Time in Mins**, ranging from 0 to 340 minutes. The horizontal axis organizes the individual tracking participants in a strict descending order from the highest duration of wasted time to the lowest. The columns are dynamically shaded using a deep crimson-to-pink gradient to emphasize the severity of the sleep loss.

Key Analytical Insights:

- **The Extreme Outlier:** The visualization highlights a massive, severe outlier on the far left of the distribution. This specific individual wastes an average of **309 minutes (over 5 hours) every single day** lying awake in bed. This represents a catastrophic sleep continuity issue that severely impacts overall wellness and daily energy levels.
- **The Secondary Risk Cluster:** Beyond the primary outlier, a secondary high-risk cluster emerges, with users losing **167 minutes (nearly 3 hours)** and **52 minutes** per day respectively. The rest of the study group stabilizes into a long tail, gradually tapering down from **44 minutes** to a highly efficient **3 minutes** of wasted time.
- **Product Strategy Implication:** This ranked distribution provides the ultimate behavioral justification for Bellabeat's marketing focus. It demonstrates that sleep vulnerability is highly individualized and skewed toward severe acute cases. Bellabeat should use these metrics to pitch the necessity of personalized, automated health alerts. For example, if the app detects a user entering the high-risk zone (for example: averaging more than 45 minutes awake in bed over a rolling 3-day period), it should automatically trigger a "Sleep Optimization Intervention"—prompting the user to log lifestyle factors in the app or offering targeted mindfulness and stress-reduction content to help minimize restlessness.

Phase 5: Share

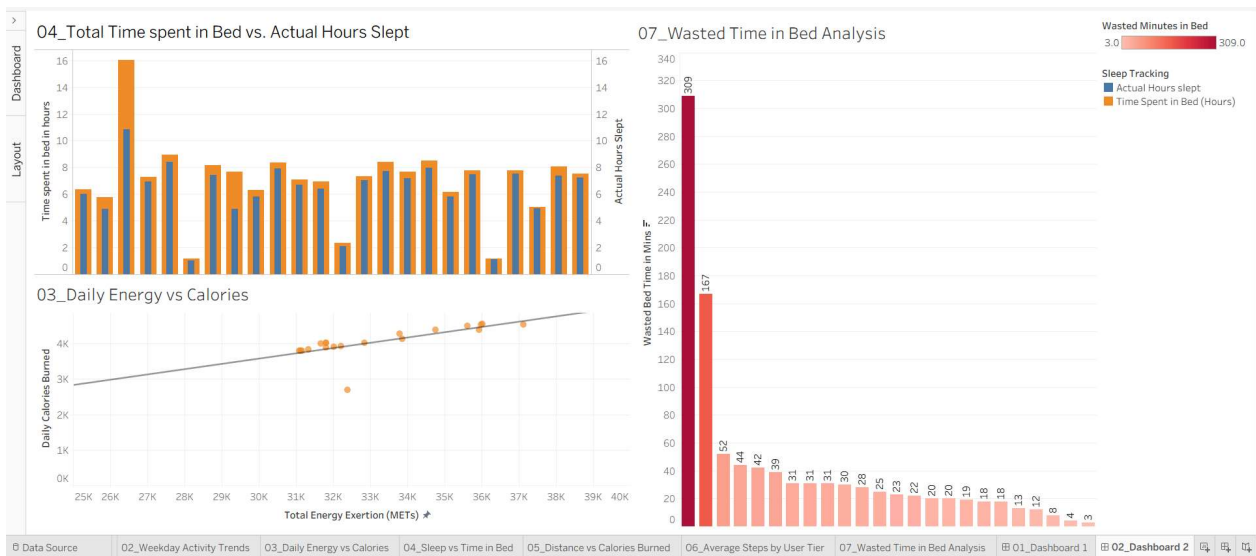
Figure 8: Activity & Engagement Dashboard (01_Dashboard 1)



Key Findings from Dashboard 1:

- **User Segment Distribution:** The consumer profile is highly diverse. Out of the sample tracked, 8 users are classified as Sedentary, 9 as Lightly Active, 9 as Fairly Active, and 7 as Very Active. Sedentary and Lightly Active users make up over half (**51.5%**) of the tracked population, representing a massive market opportunity for health-improvement tracking.
- **Average Steps by User Tier:** Performance metrics reveal a severe performance gap between behavioral tiers. *Very Active* users average an exceptional **12,487 steps** per day, easily exceeding standard clinical recommendations, whereas *Sedentary* users fall drastically behind, averaging a mere **2,936 steps** per day.
- **Weekday Activity Trends:** Group volume patterns reveal that physical activity is not uniform across the weekly cycle. **Tuesday** acts as the peak activity day of the week, reaching a scaled step index maximum of **1,207**, while activity consistently trends downward into the weekend, bottoming out on **Sunday** at an index low of **820**.
- **Distance vs. Calories Burned:** A dual-axis trend analysis shows a flawless direct correlation between distance traveled and caloric expenditure. On peak Tuesdays across the tracking period, the study group's cumulative distance total scaled to **28.5 miles**, corresponding directly with a cumulative group energy expenditure peak of **11,552 total calories burned**.

Figure 9: Sleep & Energy Dynamics Dashboard (02_Dashboard 2)



Key Findings from Dashboard 2:

- **Total Time Spent in Bed vs. Actual Hours Slept:** Thermometer chart indicators confirm a profound gap between duration in bed and actual rest. While users average a consistent baseline of **7 to 8 total hours** in bed, their actual hours slept drop significantly lower.
- **Wasted Time in Bed Analysis:** By isolating restlessness metrics, a Pareto priority distribution emerges. Sleep efficiency varies wildly by individual, with one severe outlier wasting a staggering **309 minutes (over 5 hours) awake in bed**, while a secondary cluster averages **39 to 52 minutes** wasted daily. This quantifies a significant community-wide sleep hygiene issue.
- **Daily Energy vs. Calories:** The physiological scatter plot tracks a tight cluster between **30K and 38K METs**. The clear positive slope of the regression trend line mathematically proves that daily calorie burn is reliably driven by sustained, high-intensity energy exertion (Total Energy Exertion (METs)).

Phase 6: Act

Recommendations

Based on the behavioral trends uncovered across consumer activity patterns and sleep cycles, I recommend the following targeted marketing and product development strategies for Bellabeat:

1. **Introduce an In-App "Sleep Hygiene Optimization" Module (Product Expansion):** Our analysis identifies a critical business opportunity regarding hidden sleep inefficiencies. As demonstrated in the ranked distribution, individual users waste up to **309 minutes (over 5 hours)** or an intermediate cluster of **39 to 52 minutes** daily lying awake or restless in bed. Bellabeat should market targeted software features within the *Bellabeat App*, such as automated bedroom wind-down reminders, sleep-induction soundscapes, and guided evening mindfulness sessions. This directly addresses a widespread consumer pain point using data-backed product features.
2. **Deploy "Mid-Week Momentum" Smart Notifications (Targeted Engagement):** Group tracking data shows that physical activity aggressively peaks on **Tuesdays (activity index: 1,207)** and suffers a severe slump over the weekend, bottoming out on **Sundays (activity index: 820)**. To smooth out this consistency gap, Bellabeat's *Leaf* or *Time* smart devices should leverage adaptive haptic nudges. Pushing a personalized "Weekend Warrior" or "Saturday Step-Up" incentive notification on Friday afternoons can proactively motivate users to maintain active habits when their routine loses structure.
3. **Gamify Step Tier Transitions (User Retention):** The massive performance chasm between the *Sedentary* tier (**2,936 average steps**) and the *Fairly Active* tier (**8,681 average steps**) proves that a single, rigid baseline goal (like 10,000 steps) can be intimidating to a majority of consumers. Because *Sedentary* and *Lightly Active* individuals make up **51.5%** of our tracked user segment, Bellabeat should introduce an incremental, gamified milestone progression system within the app. Rewarding users with digital badges or community recognition for graduating from Sedentary to Lightly Active scales down the intimidation factor and builds long-term product loyalty.
4. **Position the Bellabeat Membership via Predictive Exertion Metrics (Premium Monetization):** The tight linear regression model mapping *Total Energy Exertion (METs)* against *Daily Calories Burned* proves that the ecosystem's calorie-tracking metrics are highly accurate and tied directly to true physiological effort. Bellabeat should leverage this accurate benchmarking to market its subscription-based *Bellabeat Membership*. Marketing campaigns should highlight how hardware data syncs seamlessly with the membership

platform to instantly push automated, customized recovery metrics and nutritional advice the moment a user logs a high-MET workout day.

Conclusion

This comprehensive analysis demonstrates that while high-performing users maintain optimal physical habits, the vast majority of smart device consumers face prominent operational barriers regarding weekend step consistency and nighttime sleep hygiene. While users actively utilize tracking devices to log daily metrics, they clearly lack the behavioral scaffolding necessary to optimize their daily routines.

By strategically shifting marketing narratives away from passive device tracking, Bellabeat can focus on highly personalized, data-driven health optimization. Marketing campaigns should specifically target sleep efficiency, custom tier progression, and weekend engagement. This strategic pivot will allow Bellabeat to successfully capture high-potential wellness consumers globally. Ultimately, transitioning from a traditional hardware manufacturer to a holistic digital wellness partner will unlock massive new growth opportunities, securing Bellabeat's position as a premier leader in the global smart device market.